SPARKGRID: Spatiotemporal Harmonization & God's Eye AI for NASA MODIS & VIIRS Active Fire Hot Spots

Team Name: SparkGrid | Local Event: NASA Space Apps Cairo 2026
Selected Challenge: Harmonization of MODIS and VIIRS Hot Spots
Live Web & Mobile App (No Login): https://sparkgrid.masarat.app
Live 7-Slide Presentation Deck: https://sparkgrid.masarat.app/slides.html
Live FastAPI Swagger Documentation: https://sparkgrid-api.masarat.app/docs
Official Public GitHub Repository: https://github.com/freefirenoobs/sparkgrid

1. HIGH-LEVEL PROJECT SUMMARY

We built SparkGrid, an automated satellite wildfire harmonization engine and interactive 'God's Eye' AI Tactical Map addressing the 'Harmonization of MODIS and VIIRS Hot Spots' challenge.

THE PROBLEM: When wildfires break out, emergency responders rely on two NASA satellite instruments to track active blazes from space: MODIS (which scans in 1,000m x 1,000m = 1.0 km2 squares) and VIIRS (which scans in sharper 375m x 375m = 0.14 km2 squares). Plotting both raw feeds together creates severe confusion: a single fire triggers up to 8 overlapping duplicate alarms, burned areas look artificially inflated, and stationary oil/gas flares in desert and coastal basins are mistaken for real wildfires.

WHAT WE BUILT & WHY IT MATTERS: SparkGrid automatically ingests raw NASA MODIS and VIIRS satellite detections and fuses them into one deduplicated operational map. Using spherical Haversine DBSCAN spatial clustering (1.5 km radius across a 120-minute orbital window), Fire Radiative Power (FRP) weighted centroids, and Bayesian probability fusion, SparkGrid compressed 27,672 noisy raw detections into 11,505 unified fire clusters (2.45x clutter reduction) and isolated 672 Dual-Sensor Verified wildfires confirmed simultaneously by both satellites. Enriched with live Open-Meteo wind vectors across 10,106 clusters and industrial flare masking (175 flares filtered), responders can speak directly to the map ('Take me to Egypt') to fly the camera, compare true-scale 1000m vs 375m footprints, and generate 3-step AI evacuation action plans.

2. PROJECT DETAILS (HOW IT WORKS, DELIVERED FEATURES, LIMITATIONS & FUTURE

HOW IT WORKS (6-STAGE SCIENTIFIC PIPELINE IN /data-pipeline/harmonize.py):
1. Confidence Normalization: Maps MODIS integer confidence (conf / 100.0) and VIIRS categorical tiers (low=0.30, nominal=0.70, high=0.95) into a unified Bayesian probability space [0.0, 1.0].
2. Spherical Great-Circle Haversine DBSCAN: Groups raw detections on the Earth's sphere using radian Haversine distance (epsilon = 1.5 km) and an orbital overpass time window (|Delta t| <= 120 minutes).
3. FRP-Weighted Centroids: Computes cluster centers weighted by Fire Radiative Power (Center = Sum(Coord_i * FRP_i) / Sum(FRP_i)), anchoring the unified coordinates to the peak flaming front.
4. Bayesian Multi-Sensor Fusion: Computes C_fused = 1 - Prod(1 - c_i) when both MODIS and VIIRS detect the same fire, setting is_multi_sensor = true and refining resolution from 1000m to 375m.
5. Open-Meteo 10m Wind Vector Fusion: Queries 50-coordinate batches to fetch wind_speed_mps and wind_dir_deg, computing downwind flame propagation azimuth: spread_heading_deg = (wind_dir + 180) mod 360.
6. Industrial Gas Flare Masking: Flags stationary hydrocarbon flares (is_industrial = true) across oil/gas basins.

DELIVERED WORKING FEATURES & VERIFIED EMPIRICAL EVIDENCE (LIVE AT https://sparkgrid.masarat.app):
- Global + Egypt Dataset: 27,672 raw detections compressed into 11,505 unified clusters (2.45x deduplication), 672 Dual-Sensor Verified fires, 10,106 wind-enriched clusters, 175 industrial flares masked, and 296,892.71 MW total FRP.

- Egypt Case Study (207 Egypt points / 45 in top wind slice): Side-by-side 1000m MODIS (SG-EG-0003) and 375m VIIRS (SG-EG-0010, SG-EG-0011) agricultural burns in Beni Suef (29.017N, 31.204E) and Gulf of Suez oil flares (29.691N, 32.322E).
- True-Scale 1km vs 375m Footprint Polygons: Renders 1000m orange MODIS squares, 375m crimson VIIRS squares, and nested 375m cores inside 1000m dashed boxes for Dual-Verified clusters.
- Fire-Spread Azimuth Arrows & Wind Speed Heat Layer: Rotated SVG directional arrows (spread_heading_deg) plus dynamic marker sizing and 4-tier thermal color risk.
- God's Eye Voice & Camera AI Commander + Responder Action Plans: Voice/text commands fly the camera and narrate telemetry aloud via Web Speech API; POST /api/hotspots/{id}/briefing generates 3-point firefighter plans.
- Smartphone & Desktop Ready: Responsive 4-tab sidebar on desktop + slide-up bottom sheet on smartphones, 24h time-scrubber playback, and 1-click GeoJSON/CSV exports.

CURRENT LIMITATIONS & FUTURE ROADMAP:
- Limitations: Polar-orbiting satellites pass every few hours rather than continuous video, and heavy cloud cover can block mid-IR radiance. Spread arrows currently model wind vector direction/speed rather than full topographic slope perimeters.
- Future Plans: Integrate geostationary feeds (Meteosat SEVIRI / GOES) for 15-min updates, add Rothermel elliptical spread modeling with NASA SRTM slope, and add automated SMS civil defense alerts.

TOOLS, LANGUAGES, SOFTWARE & HARDWARE:
- Stack: Python 3.12, Pandas, NumPy, Scikit-Learn (Haversine DBSCAN), FastAPI, Uvicorn, React 18, Vite, Leaflet / React-Leaflet, Web Speech API.
- Hardware/Cloud: Dual Microsoft Azure Ubuntu 24.04 LTS VMs (4 GB RAM + 8 GB Swap), Cloudflare Zero Trust SSL Tunnels, and automated 2-minute GitHub CI/CD (deploy.sh).

3. NASA DATA & RESOURCES USED

1. NASA FIRMS - MODIS Collection 6.1 Near Real-Time Active Fire Product (MCD14DL):
   Link: https://firms.modaps.eosdis.nasa.gov/
   Role: Provided global (-180,-90,180,90) 24h and Egypt 3-day 1km thermal anomalies from NASA Terra and Aqua satellites (Channels 21/22 at 3.96 um, FRP, and 0-100% confidence), normalized and rendered as 1000m x 1000m footprint polygons.
2. NASA FIRMS - VIIRS 375m Active Fire Product (VNP14IMGTDL_NRT / VJ114IMGTDL_NRT / VJ214IMGTDL_NRT):
   Link: https://www.earthdata.nasa.gov/data/tools/firms
   Role: Provided 375m I-Band (3.74 um I4 & 11.45 um I5) detections from Suomi-NPP, NOAA-20, and NOAA-21. Fused with MODIS via Haversine DBSCAN and Bayesian probability to isolate 672 Dual-Sensor Verified wildfires.
3. NASA Earthdata LANCE (Land, Atmosphere Near real-time Capability for EOS):
   Link: https://www.earthdata.nasa.gov/learn/find-data/near-real-time
   Role: Provided near-real-time orbital pass timing and radiometric specifications for multi-sensor temporal alignment.

4. SPACE AGENCY PARTNER & OTHER DATA / REFERENCES

1. NOAA Joint Polar Satellite System (JPSS - Suomi-NPP, NOAA-20, NOAA-21 VIIRS): https://www.nesdis.noaa.gov/our-satellites/currently-flying/joint-polar-satellite-system (Public Domain - U.S. Gov / NOAA & NASA).
2. Open-Meteo Global Weather API: https://open-meteo.com/ - Used to enrich 10,106 clusters with 10m wind speed and direction (CC BY 4.0).
3. Esri ArcGIS Online World Imagery & Dark Gray Base Tiles:

https://www.arcgis.com/home/item.html?id=10df2279f9684e4a9f6a7f08febac2a9 (Esri Master License / Attribution on map).

4. OpenStreetMap Global Tiles: https://www.openstreetmap.org/ (ODbL License - OpenStreetMap contributors).

5. Open-Source Libraries: Scikit-Learn (https://scikit-learn.org/ - BSD-3), FastAPI (https://fastapi.tiangolo.com/ - MIT), React & React-Leaflet (https://leafletjs.com/ - MIT/BSD-2).

5. ARTIFICIAL INTELLIGENCE (AI / SI) DISCLOSURE

Yes - AI tools were utilized transparently in two capacities:

1. In-Application AI Feature (God's Eye Voice AI & Incident Briefing Engine): SparkGrid includes an interactive AI assistant (/api/ai/chat and /api/hotspots/{cluster_id}/briefing) that translates harmonized NASA MODIS/VIIRS satellite telemetry and Open-Meteo wind vectors into spoken scientific explanations, automated camera flights, and 3-point emergency responder action plans.

2. Development & Documentation Assistance: Generative AI coding assistants were used during development to accelerate React UI styling, FastAPI endpoint scaffolding, and documentation formatting. All spatio-temporal clustering algorithms (Haversine DBSCAN, FRP-weighted centroids, Bayesian fusion) and empirical benchmarks were verified directly against real NASA FIRMS datasets.